

PROJECT REPORT

SHIFTY: YOUR ROOM SHIFTING SOLUTION

AN AI-POWERED PERSONAL RELOCATION PLATFORM

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

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CANDIDATE'S DECLARATION

We hereby certify that the work which is being presented in the project report entitled **“SHIFTY: Your Room Shifting Solution – An AI-Powered Personal Relocation Platform”** in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering in the Department of Computer Science and Engineering of the Graphic Era Hill University, Dehradun, shall be carried out by the undersigned under the supervision of **Mr. Deepak Upadhayay, Assistant Professor**, Department of Computer Science and Engineering, Graphic Era Hill University, Dehradun.

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This project has been developed under the guidance and supervision of the undersigned faculty guide on the topic **“SHIFTY: Your Room Shifting Solution”**.

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ABSTRACT

The logistics of residential relocation, even for small-scale moves, are fraught with inefficiencies, opaque pricing structures, and a lack of personalised customer service. The process is often manual, time-consuming, and stressful for the user. This project, titled **SHIFTY**, presents a comprehensive mobile application designed to streamline and revolutionise the room-shifting experience by leveraging the power of Machine Learning (ML) and Artificial Intelligence (AI).

The core of SHIFTY is a data-driven approach to logistics. The application begins by systematically collecting user inventory data, allowing users to catalogue their belongings and specify items that require special handling such as delicate or heavy objects. This data serves as the input for a suite of intelligent modules. An ML-based recommendation engine analyses the inventory size, item types, and fragility to suggest an optimised service package including appropriate vehicle size, required manpower, and necessary packing materials. Concurrently, a dynamic pricing and offer system, also powered by an ML algorithm, calculates a transparent, fair, and real-time quote based on variables such as distance, inventory volume, item fragility, time of booking, and local demand.

To enhance user experience and provide round-the-clock support, an integrated AI chatbot assists users with navigation, answers queries, and facilitates the booking process. The frontend interface offers real-time system insights, including booking status, live tracking of consignments, and notification of delivery milestones. Authentication and access control mechanisms are implemented to restrict unauthorised access to sensitive user data.

This report provides a holistic overview of the project lifecycle, from initial ideation and requirement analysis to system architecture, detailed implementation of the core ML models, deployment strategies, and evaluation. It documents the design and development of the platform, showcasing how AI-driven features create a seamless, efficient, and trustworthy solution for the modern urban mover.

TABLE OF CONTENTS

Chapter No.	Section	Title	Page
		ACKNOWLEDGEMENT	i
		ABSTRACT	ii
		TABLE OF CONTENTS	iii
		LIST OF TABLES	v
		LIST OF FIGURES	vi
		ABBREVIATIONS	vii
		NOTATIONS	viii
1		Introduction	1
	1.1	Background and Motivation	1
	1.2	Problem Statement	2
	1.3	Objectives of the Project	3
	1.4	Organization of the Report	4
2		Literature Review	5
	2.1	Traditional Packers and Movers Systems	5
	2.2	Web and Mobile-Based Booking Platforms	6
	2.3	Machine Learning in Logistics	6
	2.4	Dynamic Pricing in Service Platforms	8
	2.5	Real-Time Tracking and Location Systems	8
	2.6	Conversational AI for Service Booking	9

Chapter No.	Section	Title	Page
	2.7	Comparative Study of Moving Service Approaches	9
	2.8	Conclusion of Literature Review	10
3		System Requirements and Analysis	11
	3.1	Overview	11
	3.2	Purpose	11
	3.3	Hardware Requirements	11
	3.4	Software Requirements	12
	3.5	Functional Requirements	13
	3.6	Non-Functional Requirements	14
	3.7	Feasibility Study	15
	3.8	System Analysis Summary	16
4		System Design and Architecture	17
	4.1	Overview	17
	4.2	System Architecture	17
	4.3	Components and Modules	18
	4.4	System Design Workflow	19
	4.5	Flow Diagram	19
	4.6	Design Considerations	20
	4.7	Database Schema Design	21
	4.8	Workflow of Shifty System	22

Chapter No.	Section	Title	Page
5		Implementation	23
	5.1	Overview	23
	5.2	Software Stack	23
	5.3	System Implementation	24
	5.4	Algorithm Implementation	25
	5.5	System Flow Control	26
	5.6	Performance and Optimization	26
	5.7	Testing and Validation	27
	5.8	Software Setup and Code Integration	28
	5.9	Real-Time System Monitoring and Dashboard Output	28
6		Conclusion and Future Scope	30
	6.1	Conclusion	30
	6.2	Future Enhancements	31
APPENDIX			33
REFERENCES			61

LIST OF TABLES

Table No.	Title
Table 1	Comparison of Traditional vs. AI-based Moving Platforms
Table 2	Functional Requirements of SHIFTY System
Table 3	Non-Functional Requirements of SHIFTY System
Table 4	Module-wise Implementation Status
Table 5	Testing and Validation Summary

LIST OF FIGURES

Figure No.	Title
Figure 1	Pain-point Fishbone Diagram of Traditional Relocation Experience
Figure 2	SHIFTY Target Audience and Their Needs
Figure 3	High-level System Architecture of SHIFTY
Figure 4	Entity-Relationship Diagram of the SHIFTY Database
Figure 5	UML Use-case Diagram of SHIFTY Platform
Figure 6	Data Flow of the AI-Powered Package Recommender
Figure 7	Dynamic Pricing Pipeline using XGBoost
Figure 8	Conversational Assistant Interaction Flow
Figure 9	Live Tracking Socket Architecture
Figure 10	Payment Subsystem Sequence Diagram
Figure 11	Module-wise Completion Progress Chart
Figure 12	Landing Page of the Shifty Web Prototype
Figure 13	Booking Planner Capturing Itinerary and Inventory
Figure 14	Price Prediction Screen Showing Model-Generated Quote
Figure 15	Payment Review Screen Prior to Gateway Hand-off
Figure 16	Sign-in Screen for Secure Access in Shifty System
Figure 17	Account Creation Screen with Inline Validation

ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NLP	Natural Language Processing
API	Application Programming Interface
REST	Representational State Transfer
UI	User Interface
UX	User Experience
SPA	Single Page Application
JWT	JSON Web Token
JSON	JavaScript Object Notation
HTTP	Hyper Text Transfer Protocol
HTTPS	Hyper Text Transfer Protocol Secure
DB	Database
SQL	Structured Query Language
NoSQL	Not Only SQL
CRUD	Create, Read, Update, Delete
GPS	Global Positioning System
RF	Random Forest

Abbreviation	Full Form
XGB	Extreme Gradient Boosting
MAE	Mean Absolute Error
SDK	Software Development Kit
OS	Operating System
CI/CD	Continuous Integration / Continuous Deployment
CLI	Command Line Interface
IDE	Integrated Development Environment

NOTATIONS

Symbol	Description
D_{km}	Source-to-destination distance in kilometres
V_{inv}	Total inventory volume
N_{frag}	Count of fragile items in the inventory
P_{final}	Final predicted price for a booking
P_{pack}	Recommended service package (Basic Standard Premium)
T_{move}	Scheduled move date and time
α	Demand-surge multiplier
β	Fragility penalty weight
λ	Regularisation coefficient of the XGBoost model
Q_{user}	Natural-language query from the customer
R_{bot}	Response generated by the AI chatbot